

Acceptance Testing

UAT Execution & Report Submission

Date	01 July 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	4 marks

1. Purpose of the Document

The purpose of this document is to summarize the User Acceptance Testing (UAT) activities carried out for the **Smart Agricultural Production Optimization Engine (OptiCrop)**. User Acceptance Testing was performed after completing the implementation and integration of the machine learning model, Flask backend, and web application.

The objective of UAT was to verify that the system satisfies the functional requirements, produces accurate crop recommendations, and provides a reliable and user-friendly experience for farmers and agricultural researchers before final deployment.

2. Defect Analysis

The following table summarizes the defects identified during development and testing and their final resolution status.

Resolution	Critical	Major	Minor	Cosmetic	Total
Fixed	0	4	6	3	13
By Design	0	1	2	1	4

Duplicate	0	0	1	0	1
Not Reproduced	0	0	1	0	1
Deferred	0	0	0	1	1
Total	0	5	10	5	20

Defect Summary

- No critical defects remained before release.
- All functional defects affecting crop prediction and backend processing were successfully resolved.
- Minor interface improvements and cosmetic issues were corrected during the final integration stage.
- The application successfully completed User Acceptance Testing without any blocking issues.

3. Test Case Analysis

The following table summarizes the execution status of all major functional modules.

Module	Total Test Cases	Not Tested	Failed	Passed
User Interface	8	0	0	8
Crop Recommendation Module	10	0	0	10
Crop Suitability Evaluation	6	0	0	6
Machine Learning Prediction Engine	8	0	0	8
Flask API & Backend	6	0	0	6
Dataset Analysis Module	7	0	0	7
Input Validation & Error Handling	5	0	0	5
System Integration	6	0	0	6

Total	56	0	0	56
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User Acceptance Summary

User Acceptance Testing confirmed that the Smart Agricultural Production Optimization Engine (OptiCrop) satisfies the functional and non-functional requirements defined during the Requirement Analysis and Project Design phases.

The integrated system successfully:

- Validated agricultural input parameters.
- Generated accurate crop recommendations.
- Evaluated crop suitability.
- Performed agricultural dataset analysis.
- Loaded the trained machine learning model successfully.
- Produced prediction results through the Flask web application.
- Maintained stable performance during repeated testing.

No critical issues remained at the completion of User Acceptance Testing. Based on the successful execution of all planned test cases, the system is considered **accepted for final demonstration, documentation, and deployment.**

